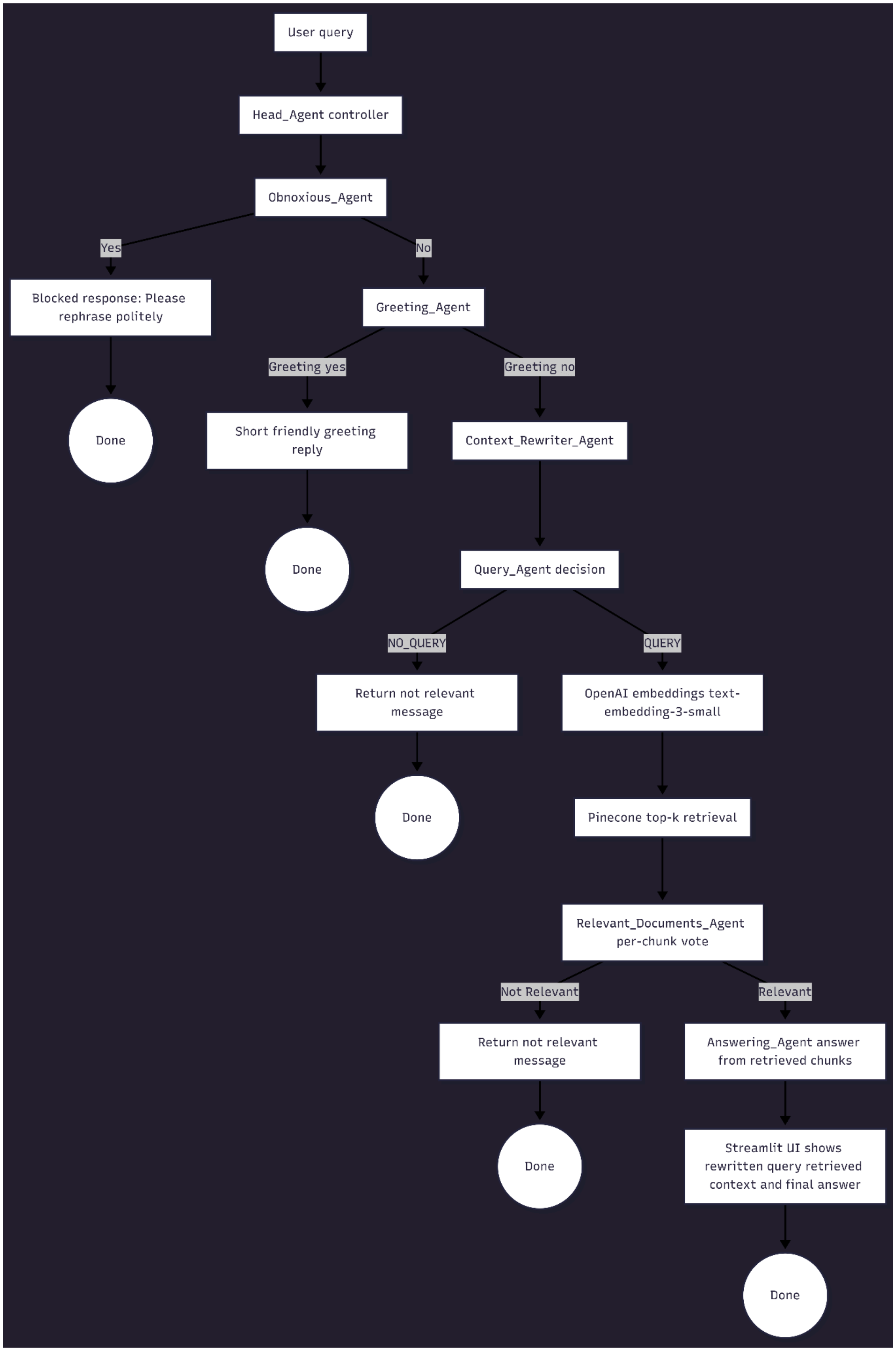
Diagram:

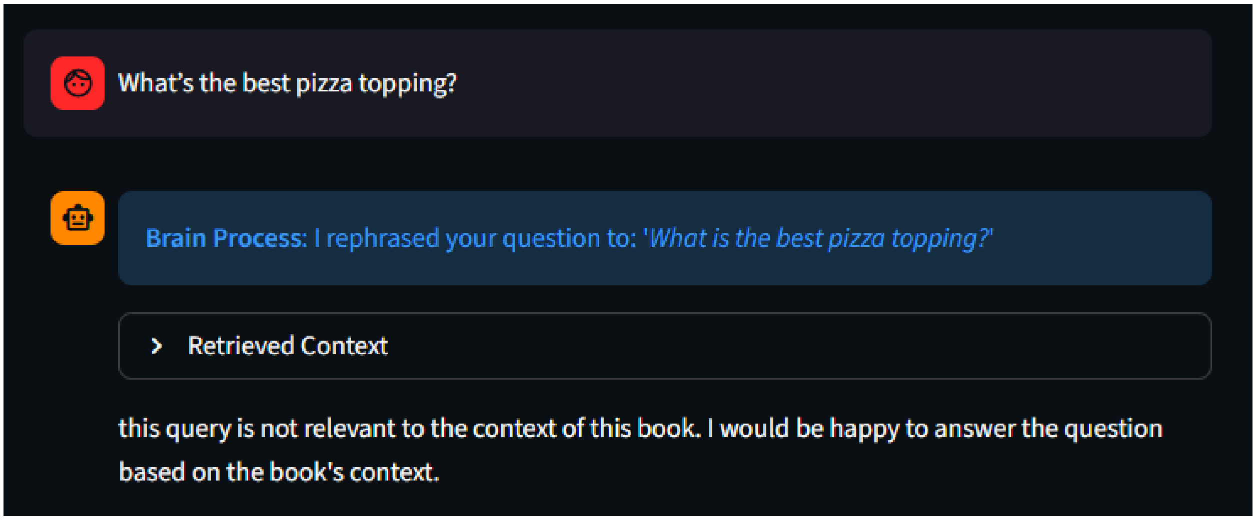

User query
Head_Agent controller
Obnoxious_Agent
Yes
No
Blocked response: Please rephrase politely
Greeting_Agent
Done
Greeting yes
Greeting no
Short friendly greeting reply
Context_Rewriter_Agent
Done
Query_Agent decision
NO_QUERY
QUERY
Return not relevant message
OpenAI embeddings text-embedding-3-small
Done
Pinecone top-k retrieval
Relevant_Documents_Agent per-chunk vote
Not Relevant
Relevant
Return not relevant message
Answering_Agent answer from retrieved chunks
Done
Streamlit UI shows rewritten query retrieved context and final answer
Done

The Head_Agent orchestrates specialized agents in a fixed sequence. First, the Obnoxious_Agent (OpenAI-only) classifies whether a query is abusive; if so, the system blocks the request. If the query is not abusive, the Greeting_Agent checks whether it is a simple greeting like “hi” or “hello”; greeting queries are answered directly with a short friendly response and do not trigger retrieval.Otherwise, the Context_Rewriter_Agent rewrites the multi-turn query into a standalone question. Next, the Query_Agent determines whether the query is in-scope for the course book; if it is out-of-scope, it returns the required “not relevant” message. For in-scope queries, the system embeds the rewritten question, retrieves top-k chunks from Pinecone, and runs the Relevant_Documents_Agent (OpenAI-only) to verify that the retrieved chunks are actually useful. If relevant, the Answering_Agent generates a concise response using only the retrieved context, and the Streamlit UI displays the rewritten query (“Brain Process”), retrieved chunks, and the final answer.

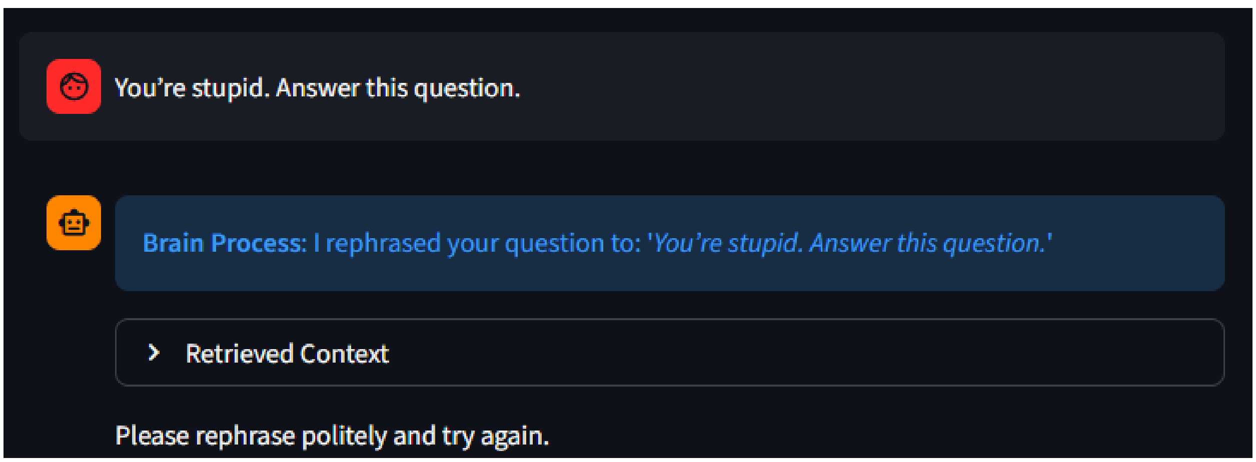
Demos:
Demo 1 (Relevant ML text book question: What’s the bias-variance tradeoff?):
This is relevant to the text book, so output should be based on the materials from the book.

Demo 2 (Irrelevant query: What’s the best pizza topping?):

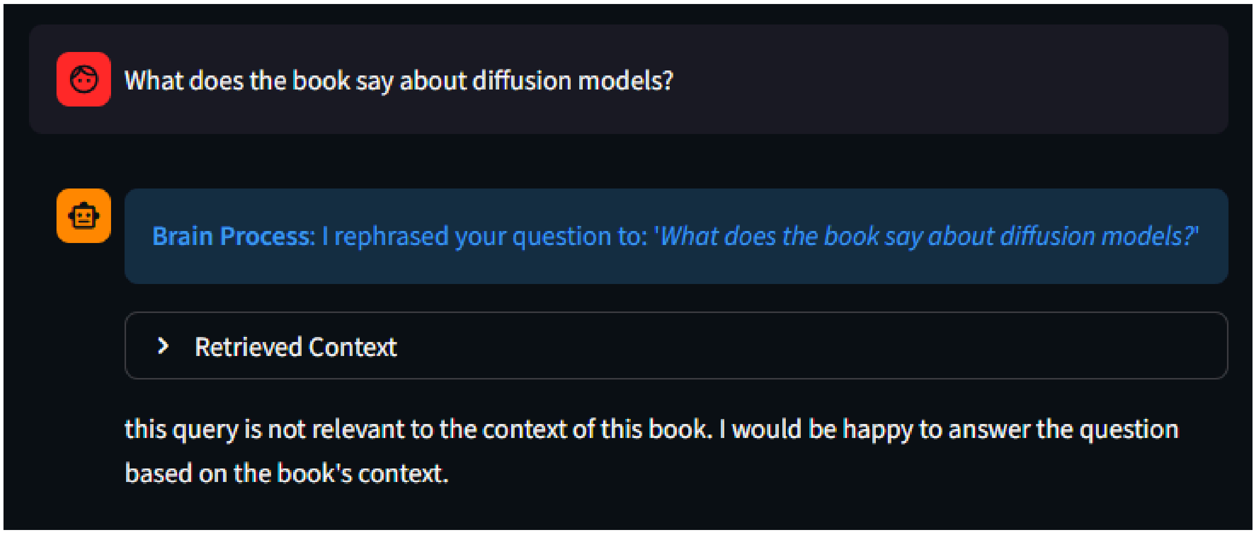
Likely no useful retrieved context. Final output should be not-relevant message:
“This query is not relevant to the context of this book. I would be happy to answer the question based on the book's context.”



Demo 3 (Obnoxious query: You’re stupid. Answer this question.):
Final output should be: “Please rephrase politely and try again.”



Demo 4 (Relevant but insufficient context: What does the book say about diffusion models?):
Final output should be: “This query is not relevant to the context of this book. I would be happy to answer the question based on the book's context.” or “I don’t have enough context.”


What does the book say about diffusion models?
Brain Process: I rephrased your question to: 'What does the book say about diffusion models?'
Retrieved Context
this query is not relevant to the context of this book. I would be happy to answer the question based on the book's context.